Sebastian Villalobos

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ML practitioner focused on mechanistic interpretability and LLM evaluation. Built an independent, architecture-agnostic interpretability toolkit and a steering-vector research pipeline on Llama-3.2-3B, including self-identified failure analysis and public null-result documentation. Production LLM evaluation and RLHF alignment experience at Scale AI. Co-authored publication in peer-reviewed journal

EDUCATION

Georgetown University — M.S. in Data Science & Analytics **Expected May 2027**

- GPA: 4.0/4.0. Teaching Assistant: Database Systems & SQL; AI Modeling
- Tuition scholarship for academic merit

Universidad Iberoamericana — B.S. Mechatronics & Production Engineering **Mexico City**

- Tuition scholarship for academic merit

RESEARCH PROJECTS

ModelLens — Neural Network Interpretability Toolkit (GitHub) **Georgetown University — In Progress**

- Built a from-scratch, architecture-agnostic PyTorch interpretability toolkit (transformers, CNNs, LSTMs, MLPs) with a unified API for activation and attribution patching, automated circuit discovery, logit lens, residual-stream analysis, dictionary-learning sparse autoencoders, and linear concept probing — 24 modules, 35 tests, also exposed as an MCP server for agent workflows
- Applied it to a circuit-level comparison of CAA steering vs. LoRA fine-tuning on Llama-3.2-3B: the circuit analysis shows the same dissociation as the behavioral instruments — LoRA rewires the output layer while CAA leaves core circuits essentially unchanged

Stoic Steering — Activation Steering & Failure Analysis (GitHub) **Personal Project — In Progress**

- Built an end-to-end steering pipeline for Llama-3.2-3B: CAA + LoRA fine-tuning, Claude-API contrastive-pair generation, layer/coefficient sweeps, and LLM-as-judge + judge-free forced-choice evaluation with seed-averaged, decoding-matched experimental controls
- Self-identified a critical decoding-asymmetry confound in the pipeline's original positive result (steered outputs sampled/truncated vs. greedy baselines). Under matched decoding, activation steering produces null effects across all three instruments; documented the full mechanism and re-measured numbers publicly

Political Sentiment Analysis (GitHub) **Georgetown University**

- Developed DistilBERT-based NLP pipeline for emotion classification across 2,000+ Congressional speeches, identifying statistically significant partisan patterns

WORK EXPERIENCE

Welo Data — Data Science & Analytics Engineer **Nov 2024 – Aug 2025**

- Developed XGBoost and GMM models across 120k+ records for task-difficulty prediction and rater-project matching, improving annotation accuracy by 10–15%
- Automated financial ETL pipeline in Redshift processing high-volume transactions (0.001% error rate), reducing reconciliation time by 70%

Scale AI — Data Operations Engineer **Oct 2022 – Nov 2024**

- Developed automated evaluation frameworks for NVIDIA Nemotron's RLHF alignment using GPT-4 as Judge-LLM and HelpSteer2 methodology
- Built LLM quality linting system adopted across GenAI organization, automating error detection via programmatic flagging and reducing manual review by 40%
- Built TypeScript-MongoDB integrations and Python pipelines for end-to-end data processing

Scale AI — Operations Data Analyst & Specialist **Feb 2021 – Oct 2022**

- Built logistic regression model on 12+ months of operational data predicting batch acceptance, improving approval rates by 20%

PUBLICATION

Villalobos-Alva Jalil, ..., Villalobos-Alva Sebastian, et al. (2022). *Protein Science Meets Artificial Intelligence: A Systematic Review*. Front. Bioeng. Biotechnol.

SKILLS & CERTIFICATIONS

Interpretability & AI Safety: Mechanistic Interpretability (Activation & Attribution Patching, Circuit Discovery, Logit Lens, Residual Stream, SAEs, Concept Probing), LLM Evaluation (Judge-LLM, Forced-Choice), CAA, LoRA Fine-tuning, RLHF (HelpSteer2)
Programming: Python (PyTorch, TensorFlow, HuggingFace, Pandas, NumPy, Scikit-learn), SQL, NoSQL, TypeScript, R, Git/GitHub, Linux/CLI, Docker
Languages & Certifications: Spanish (Native), English (Fluent); MIT MicroMasters in Statistics and Data Science